Journal Paper 2 Summary (25 points total)

Paper Title	Human-centered Information Visualization Adaptation Engine
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1. What do you think the paper is about in layman's terms? Summarize each of the three results of the evaluations presented in Section 6 of the paper. Do you agree with the conclusions? Why or why not? [5 points]

Focus of this question
• This question encourages you to evaluate the arguments, evidence, assumptions, and conclusions about key issues (i.e. think critically about the paper) • This question encourages you to develop your own knowledge, comprehension and conceptual understanding and to connect, synthesize, and/or transform your ideas into a new form (i.e. be a creative thinker and contribute your ideas and thoughts)

Write your answer below:

This paper aims use an adaptation engine to make data visualizations more comprehensible by making personalized content (visualization types and adaptation elements) to hopefully improve accuracy and time-to-action efficiency. To do this, the adaptation engine is given a user, an analysis task, and two sets of adaptation rules to take into consideration before returning a "winner" visualization that suits the analyst. For the first evaluation, the performance of the participants was being measured in terms of milliseconds to address an analysis task and complete it. The results of this first task were concluded to be positive affected. The second evaluation looked at analysis accuracy, in which the participants' ability to address analysis tasks correctly was measured. The conclusion of this second evaluation was that when the adaptation engine was enabled, the participants' accuracy was positively affected. In the third evaluation, the participants were asked to response to a User Experience Questionnaire. The conclusion of this third and final evaluation was that when enabling data visualization adaptation, the user experience increased.

Overall, I think these conclusions are valid, and the researcher's way of evaluating this are sound. I agree with how the study conducted their empirical methods and their evaluation methods. Even without data, when I just first read the abstract, already I could understand how personalizing data visualizations might help people glean more information and understand what they are looking at better.

2. The authors present their view of the limitations of their paper. If you had the task of extending the author's paper with new research, state and discuss two (2) more questions about their study NOT mentioned in the paper that you would want to cover. [5 points]

Focus of this question
• This question encourages you to connect your learning to "real-world" issues or life experiences and consider diverse perspectives for the application of concepts in the paper to the real-world • This question encourages you to develop your own knowledge, comprehension and conceptual understanding and to connect, synthesize, and/or transform your ideas into a new form (i.e. be a creative thinker and contribute your ideas and thoughts)

Write your answer below:

Some of the limitations the authors stated include: the tasks that were used were not balanced in terms of task type, and the focus was usually on more simple comparison tasks, and "homogenous weak learner assumption" where they treated every rule as important.

Something that I would want covered in a future study is because these rules were all equally important, would weighting rules help find which adaptation rules are contributing more to the participants' success? And would eliminating the weaker ones give a different result?

Another question that I have is: if given more complex tasks, would the adaptation engine hold up and would there be enough data to test it?

3. The authors present three (3) questions that the design of a user adaptive system involves on page 32. State these three questions, summarize how they are addressed (as discussed in the paper) and discuss three real-world applications of how to adapt data visualizations that are NOT mentioned in the paper. [15 points]

Focus of this question
• This question encourages you to connect your learning to "real world" issues or life experiences and consider diverse perspectives for the application of concepts in the paper to the real-world • This question encourages you to reflect on what you are learning • This question encourages you to contribute your ideas and thoughts • Consider diverse perspectives (gender, political, ethnic, racial, etc.) relevant to the course and considered during class or in your assignments • Challenges you to develop and present your own knowledge, comprehension, and conceptual understanding

Write your answer below:
The three questions to designing a user adaptive system are: what to adapt to, when to adapt, and how to adapt.

The paper states that their work focuses on "what to adapt to" and "how to adapt". Regarding "what to adapt to", the study states that how a user interacts, understands, and utilizes data visualizations are based on individual differences that influence how much a user gleans. With "how to adapt", the study aimed to build a flexible "human-centered by-design adaptation engine" that uses a "multidimensional human-centered user model for delivering the best fit data visualization".

Disclaimer: I just looked up "how to adapt data visualizations" in google, and just looked at a bunch of articles so I hope I am understanding the question fully.

While this study is highly involved with human cognitive abilities to drive how they approached adaptations, I am thinking more broadly in terms of how to adapt data visualizations:

1. Accessibility adaptations: I learned this one in a graphic novel. But designing charts to be readable and interpretable for the color-blind, the dyslexic, or for screen-reader-compatibility is something to consider.
2. Layout adaptations: Displaying a lot of data usually in a dashboard or some high-level overview can be overwhelming when not considering where to place each visualization accordingly. Making an adaptation engine to check which arrangement of visualization is most appealing to a user might be helpful.

3. Interaction-based adaptations: For most people (especially non-data analysts), visualizations are meant to be kept free of clutter. Although eliminating clutter can be challenging, especially when you believe every metric to be important. Instead of choosing what chart goes best, the adaptation engine could adjust how much detail is being shown in the visualization based on user behavior.